



Technical, Architecture, Evidence, Business, and Regulatory Report

Phase 2 hackathon package | Version 0.3.0 | District
emergency-department simulation | 25 August 2026

**Safety-constrained dynamic queue coordination - not an ERP, diagnosis
system, or replacement for clinical judgement.**

Prepared for evaluation with synthetic patients only. All capacity, staffing, cost, and performance values are illustrative or machine-specific and must be replaced or independently validated before any pilot.

01 Executive decision summary

Build PatientTriage.ai as a focused emergency coordination layer: safety rules first, CDM context second, uncertainty always visible, and clinicians always in control.

The upgraded solution is a complete, testable district-ED demonstration. It shows an 18-bed operational map, patient-wise queue, five role views, deterioration and surge behavior, minimum-necessary pharmacy/blood readiness, six scheduling baselines, scalability evidence, and audited override. The BTL ranker has been replaced by a feature-conditioned CDM.

The evidence supports three claims:

- The software behaves as specified in synthetic scenarios, including edge and denial paths.
- Safety-first ordering reduces safety-weighted delay relative to arrival- or service-time-only baselines in the supplied snapshot.
- The main innovation is dynamic, uncertainty-aware, governable coordination - not a claim of clinical superiority.

Decisions:

- Keep the product outside ERP scope.
- Do not add blockchain to the MVP; retain the hash-chain audit and revisit only for a multi-organization consortium.
- Proceed next to a governed 12-week shadow-pilot design, not live autonomous use.

02 Problem, users, and product boundary

Emergency queues change after initial triage. New cases arrive, observations become stale, patients deteriorate, wait-related risk grows, and surge pressure competes with fairness. A static score cannot represent this changing operating state.

The product coordinates four decisions:

Decision	User	Output	Explicit exclusion
Who needs review next?	Nurse/doctor	Safe dynamic queue	No diagnosis or treatment
What is happening across 18 spaces?	Nurse/doctor/admin	Operational projection	No bed-master authority
What support readiness is needed?	Pharmacy/blood bank	Minimum signal + acknowledgement	No inventory/order/dispensing
Is the process controlled?	Doctor/admin	Audit, override, benchmark	No compliance claim

Non-ERP boundary: no billing, procurement, stock ledger, dispensing, cross-match management, payroll, ADT source of truth, or authoritative bed management.

03 District-hospital simulation

The active configuration represents an illustrative district-level hospital so the demo has concrete capacity, shift, and staffing constraints. It is a simulation fixture, not a standard.

Parameter	Value	Parameter	Value
Catchment	500,000	Hospital beds	200
ED spaces	18	Treatment teams	4
Shift	08:00-20:00	Normal / surge arrivals	20 / 60
Zones	3 resus, 9 acute, 6 observation	Clinical staff	3 physicians, 4 residents, 14 nurses

Localization gates:

- Replace capacity and roster with approved hospital data.
- Fit arrival distributions and reassessment intervals to local evidence.
- Map authoritative bed and observation identifiers through governed integration.
- Have clinical leadership approve every rule and escalation path.

04 Role dashboards and least privilege

Role	Reads	Writes	Dashboard emphasis
Nurse	Facility, patients, queue, beds	Intake, vitals, bed workflow	Bed-wise + patient-wise
Doctor	Queue, beds, readiness, audit	Vitals, override, disposition	Clinical control
Pharmacy	Facility, pharmacy readiness	Acknowledge	Minimum necessary
Administration	Facility, beds, analytics, audit	Scenario control	Capacity + evidence
Blood bank	Facility, blood readiness	Acknowledge	Minimum necessary

The dashboard's role header is a demonstration of authorization boundaries, not authentication. Production needs hospital SSO/OIDC, identity proofing, session policy, device/network controls, least privilege, and periodic access review.

05 18-bed and patient-wise experience

The bed board renders six columns by three rows. Available spaces are green; occupied spaces use text plus critical red, emergent orange, urgent amber, less-urgent blue, or non-urgent slate. Select a bed to open its zone, patient ID, queue position, wait, acuity, and monitoring state.

Color	Meaning	Secondary cue
Green	Available	Bed ID + available text
Red	Critical	Acuity and critical-escalation text

Color	Meaning	Secondary cue
Orange	Emergent	Acuity text
Amber	Urgent	Acuity text
Blue	Less urgent	Acuity text
Slate	Non-urgent	Acuity text

The bed board is a stable operational read model. It does not allocate, transfer, admit, discharge, or replace the hospital's bed system.

06 System architecture and data flow

Layer	Current component	Production adapter
Experience	Streamlit role dashboard	Accessible clinical web app
API	FastAPI + Pydantic	Gateway, OIDC, rate limit, policy enforcement
Decision	Rules + urgency + CDM + confidence	Versioned stateless service
Operations	Bed/readiness projection	Event-driven integration read models
Patients	Thread-safe in-memory repository	Governed transactional/FHIR adapter
Audit	Hash-chained SQLite	Managed append-only store + immutable archive

Core request path:

- Validate role and input schema.
- Read the current synthetic waiting set.
- Apply safety, patient-level urgency, contextual utility, monitoring, and confidence.
- Apply valid clinician overrides and write the event chain.
- Return a typed immutable snapshot with reasons and warnings.

07 CDM, safety, uncertainty, and failure

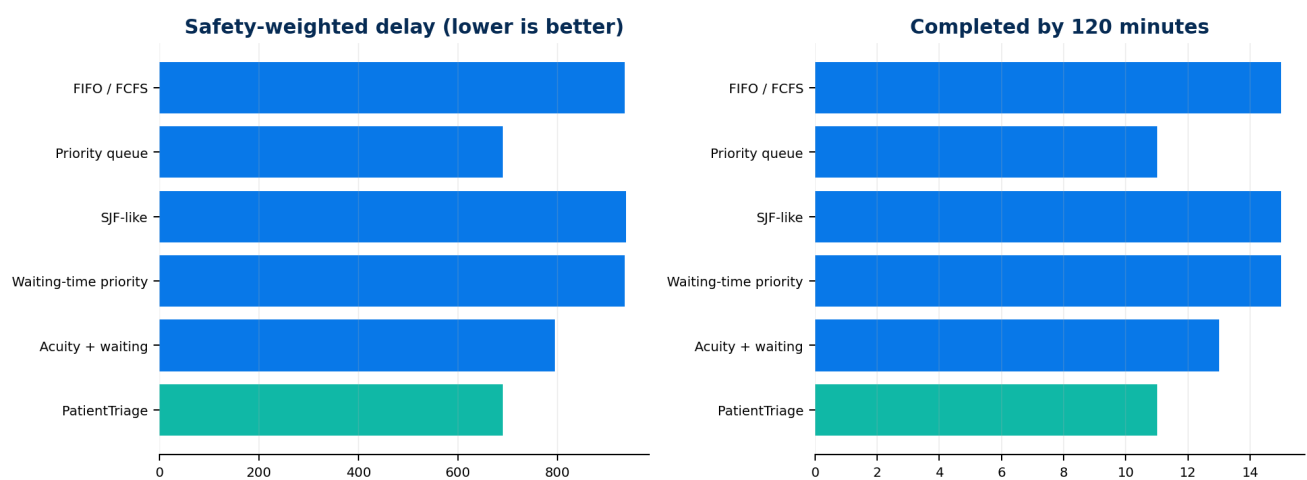
The CDM uses seven bounded features: physiology, symptoms, pain, waiting, deterioration, uncertainty, and age-group vulnerability. It adds patient-to-waiting-set interactions to patient-only utility.

Safety is lexicographic: final acuity first, CDM utility only inside the same acuity, then arrival time and deterministic ID.

Safe failure path:

- Recompute safety floors and patient-level urgency for current patients.
- Retain new arrivals rather than replaying a stale list.
- Use last-known-good order only as a within-acuity tie-break where possible.
- Mark all affected recommendations stale and low confidence.
- Request manual reassessment and keep critical alerts visible.

08 Baseline evaluation

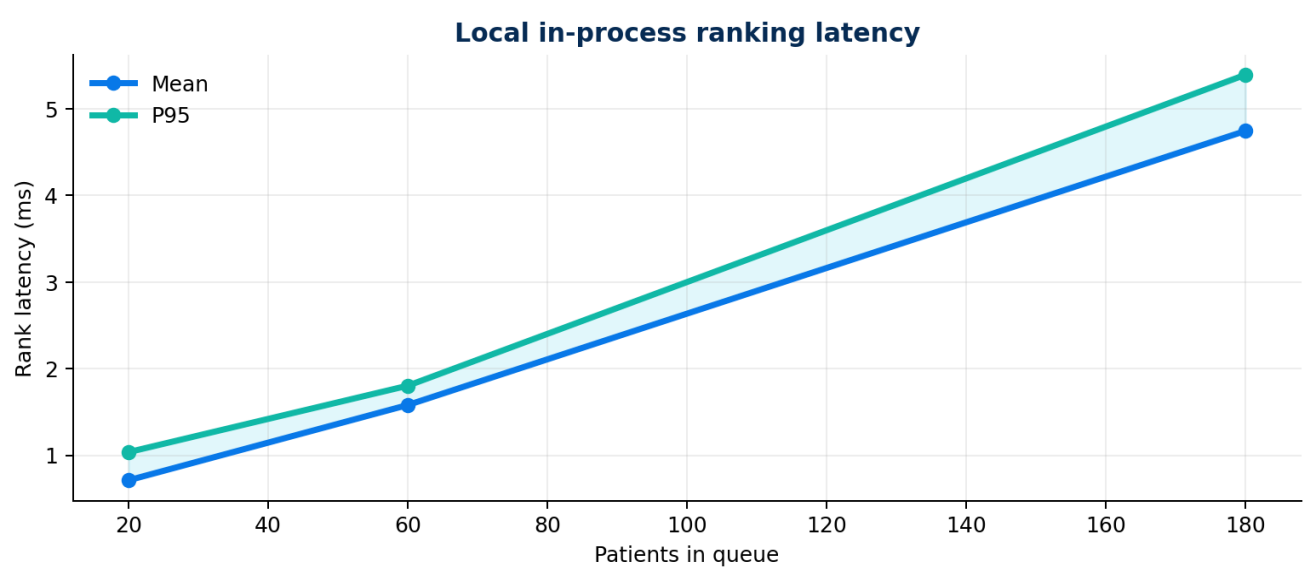


Normal 20-patient scenario, four non-preemptive treatment teams, 120-minute horizon. PatientTriage.ai ties acuity priority on safety-weighted delay because safety-first order dominates this static snapshot. SJF-like completes more cases in the horizon but delays critical work.

Policy	What it optimizes	Observed trade-off
FCFS	Arrival fairness	Critical mean added wait 110.5 min
Acuity	Clinical severity	Lowest safety delay; lower throughput
SJF-like	Short jobs	15 completed; critical delay 115 min
Waiting	Anti-starvation	Same order as FCFS in this fixture
Acuity + wait	Mixed objective	Middle safety/throughput result
PatientTriage	Safety + context + monitoring	Static tie plus dynamic governance features

This is operational evidence, not clinical validation. The strongest defensible claim is a complete dynamic and governable workflow, not universal numerical superiority.

09 Scalability and production path



Patients	Mean	P95	Scope
20	0.711 ms	1.036 ms	Normal demo
60	1.580 ms	1.803 ms	3x surge
180	4.745 ms	5.394 ms	Scale probe

These are local in-process measurements after warm-up and exclude network, persistence, identity, concurrent users, and telemetry. Production evolution: stateless replicas, managed PostgreSQL, durable event ingestion, centralized immutable logs, OIDC, secrets, observability, backups, and tested recovery.

10 Security, privacy, and audit

Prototype protections:

- Synthetic identifiers only; no names, addresses, phone numbers, or government IDs.
- Pydantic validation for ranges, formats, and timezone-aware timelines.
- Least-privilege API permissions by role.
- Mandatory override reason and versioned audit events.
- Hash chain detects event modification.

Production minimums:

- Hospital SSO/OIDC, MFA where appropriate, short sessions, and access review.
- Encryption, key management, retention/deletion policy, backup, and recovery exercises.
- Threat modeling, dependency/vulnerability management, penetration testing, and incident response.
- Provenance, idempotency, source/ingestion timestamps, and site-specific data minimization.

11 Regulatory planning map

Market	Primary linkage	Engineering consequence
US	HIPAA Security Rule where applicable; FDA CDS guidance and classification analysis	Safeguards, intended-use control, human review, device determination
UK	UK GDPR health data; MHRA SaMD; NHS DCB0129/DCB0160; DSPT	Privacy impact, clinical safety case, deployment acceptance, cyber assurance
EU	GDPR; MDR/MDCG software classification; EU AI Act risk framework	Qualification/classification, risk management, data governance, oversight, monitoring

Regulatory classification follows intended use, claims, users, and deployment. The current package is a synthetic research prototype and makes no compliance or approval claim.

Before a real pilot: classification opinion, hazard log, requirements traceability, human-factors study, data provenance/subgroup evaluation, security evidence, privacy impact assessment, downtime procedure, and independent shadow-mode protocol.

12 Blockchain and consensus

Decision: no blockchain in the MVP.

The ranking and bedside workflow do not benefit from a ledger. The current hash chain is suitable for a single-hospital prototype. Consider a permissioned ledger only when independent institutions must share audit anchors and cannot accept one operator. Never put directly identifying health data on-chain.

Option	Fault model	Fit
Hash chain	Detects rewriting when monitored/anchored	Recommended now
Fabric Raft	Crash-fault tolerant	Trusted consortium with known members
Fabric SmartBFT	Byzantine-fault tolerant	Multi-owner threat model; added overhead
Public PoW/PoS	Open adversarial membership	Reject for privacy, governance, cost, latency

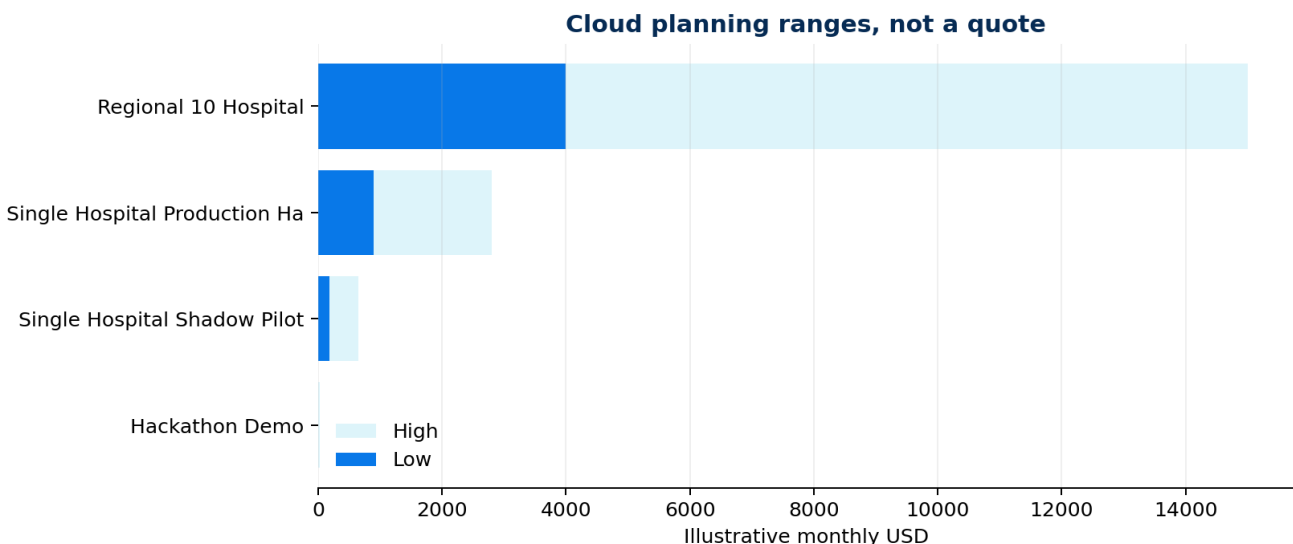
13 Business proposal and pilot

Target buyers are district/state hospitals, hospital networks, emergency clinical leadership, and quality/safety teams. The commercial path is a fixed discovery and shadow pilot, followed only after evidence gates by an annual site subscription and optional managed support.

Weeks	Work	Exit evidence
1-2	Workflow, intended use, hazards, local parameters	Signed scope and safety owner
3-5	Identity, interface, environment, localization	Authorization and integration evidence
6-8	Silent shadow operation	No effect on live care
9-10	Human factors, load, downtime, security	Exercises passed; findings triaged
11-12	Independent evaluation and go/no-go	Governance decision

Stop criteria include unsafe ranking, stale-data blindness, excessive false alerts, identity failure, or inability to recover to manual workflow. Success measures include high-risk review delay, overdue reassessment, alert acknowledgement, overrides, latency, recovery, usability, and subgroup/missing-data behavior.

14 Techno-economics



Illustrative cloud ranges use USD 1 = INR 96 on 25 August 2026. A small shadow pilot is planned at \$180-\$650 per month; a single-site high-availability production case at \$900-\$2,800. Use a current regional cloud calculator before procurement.

Pilot workstream	INR lakh range
Clinical safety and governance	5-12
Product and human factors	3-6
Backend, identity, integration	8-16
Security and compliance	4-10
Verification and validation	4-8
Training/change/contingency	4-10
Total one-time pilot delivery	28-62

Ranges exclude taxes, clinical devices, EHR vendor/interface fees, legal advice, migration, and 24x7 staffing. They are not a quote, budget approval, savings estimate, or ROI promise.

15 SWOT and strategic response

Strengths	Weaknesses
Safety floors, dynamic context, override, failure fallback	Illustrative model; no clinical validation
Five least-privilege views and clear non-ERP scope	Prototype state and demo identity are process-local
Reproducible baselines, scale data, and tests	Synthetic labels cannot prove efficacy/fairness

Opportunities	Threats
Governed district-hospital shadow pilot	Regulatory scope expansion and liability
FHIR/EHR adapter without replacing source systems	Automation bias, alert fatigue, poor data quality
Multi-site operational learning	Cyberattack, downtime, vendor/integration lock-in

Strategic response: keep scope narrow, run shadow mode first, publish limitations, require clinician control, localize parameters through governance, and treat integration, security, clinical validation, and human factors as first-class product work.

16 Verification and edge cases

159 automated tests passed with 96.19% coverage at the implementation checkpoint.

The suite covers:

- Empty, single, tied, and large synthetic queues; probability normalization and numerical stability.
- Safety floors, pediatric/geriatric boundaries, zero history, ambiguity, missing/stale data, future/out-of-order times.
- Deterioration, normal and 3x surge, model failure, new arrivals during fallback, and override safety conflicts.

- Hash integrity and tamper detection, role denial paths, malformed hospital profiles, and API validation.
- Empty/normal/surge bed boards, pharmacy/blood acknowledgement, all six baselines, and scale validation.

The final package also runs Ruff, formatting check, dependency check, wheel build, API smoke tests, Streamlit runtime checks, PDF/PPTX rendering inspection, and archive verification.

17 Limitations and next decision

- Thresholds and coefficients are illustrative and not clinically learned or approved.
- Confidence is a transparent data-quality heuristic, not outcome probability.
- Synthetic expected labels are not medical ground truth.
- The benchmark is non-preemptive and uses illustrative service times.
- No live EHR, device, lab, pharmacy, blood-bank, ADT, or identity integration is included.
- Passing software tests does not establish clinical safety, fairness, effectiveness, or compliance.

Recommended next decision: approve a scoped, independently governed shadow-pilot design - not live clinical deployment.

18 Primary references

- Indian Public Health Standards portal - <https://iphs.mohfw.gov.in/>
- MoHFW major trauma standard treatment guidance - <https://clinicalestablishments.mohfw.gov.in/sites/default/files/standard-treatment-guidelines/9451.pdf>
- NHS England model emergency department - <https://www.england.nhs.uk/long-read/the-model-emergency-department-high-performing-urgent-and-emergency-care-pathways/>
- US HHS HIPAA Security Rule - <https://www.hhs.gov/hipaa/for-professionals/security/laws-regulations/index.html>
- US FDA Clinical Decision Support Software - <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/clinical-decision-support-software>
- UK MHRA software and AI as a medical device - <https://www.gov.uk/government/publications/software-and-artificial-intelligence-ai-as-a-medical-device/software-and-artificial-intelligence-ai-as-a-medical-device>
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- Hyperledger Fabric ordering and BFT documentation - <https://hyperledger-fabric.readthedocs.io/>
- AWS Fargate, RDS PostgreSQL, ElastiCache, and CloudWatch pricing - <https://aws.amazon.com/pricing/>
- RBI reference-rate data - https://rbi.org.in/Scripts/BS_NSDPDisplay.aspx?param=4

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